

Interacting Multipoles of Rank 2 and 1:

$$T_{\alpha\beta\gamma}\Theta_{\alpha\beta}^A\mu_\gamma^B = R_z^{-7} \cdot -\frac{1}{3} \cdot \begin{bmatrix} -15 \cdot R_\alpha \cdot R_\beta \cdot R_\gamma \cdot \Theta_{\alpha\beta}^A \cdot \mu_\gamma^B \\ +3 \cdot R_z^2 \cdot R_\alpha \cdot \delta(\beta, \gamma) \cdot \Theta_{\alpha\beta}^A \cdot \mu_\gamma^B \\ +3 \cdot R_z^2 \cdot R_\beta \cdot \delta(\alpha, \gamma) \cdot \Theta_{\alpha\beta}^A \cdot \mu_\gamma^B \\ +3 \cdot R_z^2 \cdot R_\gamma \cdot \delta(\alpha, \beta) \cdot \Theta_{\alpha\beta}^A \cdot \mu_\gamma^B \end{bmatrix} \quad (1)$$

Simplify deltas:

$$T_{\alpha\beta\gamma}\Theta_{\alpha\beta}^A\mu_\gamma^B = R_z^{-7} \cdot -\frac{1}{3} \cdot \begin{bmatrix} -15 \cdot R_\alpha \cdot R_\beta \cdot R_\gamma \cdot \Theta_{\alpha\beta}^A \cdot \mu_\gamma^B \\ +3 \cdot R_z^2 \cdot R_\alpha \cdot \delta(\beta, \beta) \cdot \Theta_{\alpha\beta}^A \cdot \mu_\beta^B \\ +3 \cdot R_z^2 \cdot R_\beta \cdot \delta(\alpha, \alpha) \cdot \Theta_{\alpha\beta}^A \cdot \mu_\alpha^B \\ +3 \cdot R_z^2 \cdot R_\gamma \cdot \delta(\alpha, \alpha) \cdot \Theta_{\alpha\alpha}^A \cdot \mu_\gamma^B \end{bmatrix} \quad (2a)$$

$$\xrightarrow{\text{cleaned up}} R_z^{-7} \cdot -\frac{1}{3} \cdot \begin{bmatrix} -15 \cdot R_\alpha \cdot R_\beta \cdot R_\gamma \cdot \Theta_{\alpha\beta}^A \cdot \mu_\gamma^B \\ +3 \cdot R_z^2 \cdot R_\alpha \cdot \Theta_{\alpha\beta}^A \cdot \mu_\beta^B \\ +3 \cdot R_z^2 \cdot R_\beta \cdot \Theta_{\alpha\beta}^A \cdot \mu_\alpha^B \\ +3 \cdot R_z^2 \cdot R_\gamma \cdot \Theta_{\alpha\alpha}^A \cdot \mu_\gamma^B \end{bmatrix} \quad (2b)$$

$$\xrightarrow{\text{reduced indices}} R_z^{-7} \cdot -\frac{1}{3} \cdot \begin{bmatrix} -15 \cdot R_\alpha \cdot R_\beta \cdot R_\gamma \cdot \Theta_{\alpha\beta}^A \cdot \mu_\gamma^B \\ +3 \cdot R_z^2 \cdot R_\alpha \cdot \Theta_{\alpha\beta}^A \cdot \mu_\beta^B \\ +3 \cdot R_z^2 \cdot R_\beta \cdot \Theta_{\alpha\beta}^A \cdot \mu_\alpha^B \\ +3 \cdot R_z^2 \cdot R_\beta \cdot \Theta_{\alpha\alpha}^A \cdot \mu_\beta^B \end{bmatrix} \quad (2c)$$

$$\xrightarrow{\text{added up}} R_z^{-7} \cdot -\frac{1}{3} \cdot \begin{bmatrix} -15 \cdot R_\alpha \cdot R_\beta \cdot R_\gamma \cdot \Theta_{\alpha\beta}^A \cdot \mu_\gamma^B \\ +3 \cdot R_z^2 \cdot R_\alpha \cdot \Theta_{\alpha\beta}^A \cdot \mu_\beta^B \\ +3 \cdot R_z^2 \cdot R_\beta \cdot \Theta_{\alpha\beta}^A \cdot \mu_\alpha^B \\ +3 \cdot R_z^2 \cdot R_\beta \cdot \Theta_{\alpha\alpha}^A \cdot \mu_\beta^B \end{bmatrix} \quad (2d)$$

Simplify R:

$$T_{\alpha\beta\gamma}\Theta_{\alpha\beta}^A\mu_\gamma^B = R_z^{-7} \cdot -\frac{1}{3} \cdot \begin{bmatrix} -15 \cdot R_z \cdot R_z \cdot R_z \cdot \Theta_{zz}^A \cdot \mu_z^B \\ +3 \cdot R_z^2 \cdot R_z \cdot \Theta_{z\beta}^A \cdot \mu_\beta^B \\ +3 \cdot R_z^2 \cdot R_z \cdot \Theta_{z\alpha}^A \cdot \mu_\alpha^B \\ +3 \cdot R_z^2 \cdot R_z \cdot \Theta_{\alpha\alpha}^A \cdot \mu_z^B \end{bmatrix} \quad (3a)$$

$$\xrightarrow{\text{cleaned up}} R_z^{-7} \cdot -\frac{1}{3} \cdot \begin{bmatrix} +3 \cdot R_z^3 \cdot \Theta_{z\beta}^A \cdot \mu_\beta^B \\ +3 \cdot R_z^3 \cdot \Theta_{z\alpha}^A \cdot \mu_\alpha^B \end{bmatrix} \quad (3b)$$

$$\xrightarrow{\text{reduced indices}} R_z^{-7} \cdot -\frac{1}{3} \cdot \begin{bmatrix} +3 \cdot R_z^3 \cdot \Theta_{z\alpha}^A \cdot \mu_\alpha^B \\ +3 \cdot R_z^3 \cdot \Theta_{z\alpha}^A \cdot \mu_\alpha^B \end{bmatrix} \quad (3c)$$

$$\xrightarrow{\text{added up}} R_z^{-7} \cdot -\frac{1}{3} \cdot \begin{bmatrix} +6 \cdot R_z^3 \cdot \Theta_{z\alpha}^A \cdot \mu_\alpha^B \end{bmatrix} \quad (3d)$$

Combining everything:

$$T_{\alpha\beta\gamma}\Theta_{\alpha\beta}^A\mu_\gamma^B = \begin{bmatrix} -2 \cdot R_z^{-4} \cdot \Theta_{z\alpha}^A \cdot \mu_\alpha^B \end{bmatrix} \quad (4)$$

Simplify Dipoles:

$$T_{\alpha\beta\gamma}\Theta_{\alpha\beta}^A\mu_\gamma^B = \begin{bmatrix} -2 \cdot R_z^{-4} \cdot \Theta_{yz}^A \cdot \mu_y^B \end{bmatrix} \quad (5a)$$

$$\xrightarrow{\text{cleaned up}} [+0] \quad (5b)$$

Combining everything:

$$T_{\alpha\beta\gamma}\Theta_{\alpha\beta}^A\mu_{\gamma}^B = +0 \tag{6}$$